

Emotion understanding in infants and young children: How input shapes emotional development

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Abstract

Here, we will review the developmental literature on how infants and young children learn about emotions. We take a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. We will also emphasize the role of *input* into this development, a factor that has often been ignored. Indeed, little research has systematically sought to understand how children's emotional development is shaped by the emotional input they receive. We focus on the development of three abilities that comprise emotion understanding. First, we will discuss the developmental trajectory of *emotional expression*, and how infants and young children come to express emotions and behave emotionally. Second, we will discuss how infants and young children learn to *regulate* those emotional responses, and how different environmental inputs affect that ability. Third, we will discuss the development of *emotion perception*, or how infants and young children learn how to perceive, categorize, and identify the emotions of others over time. Finally, we will discuss mechanisms for developmental change in all of these domains, and how other concurrent domains of development—including motor skill, language, theory of mind, and attachment—might have cascading effects on changes in emotion understanding.

Keywords: Emotion perception, Emotion regulation, Emotion understanding, Emotional expression, Emotional facial configurations, Emotion, Input

Over the course of a day, adults engage in an average of about 12 social interactions with various individuals, including friends, colleagues, teachers, children, and strangers (Zhaoyang et al., 2018). This suggests that over the course of our lifetimes, we will likely participate in hundreds of thousands of social interactions, spending a significant part of our lives engaged with others. Emotion understanding—or the set of abilities related to determining our own emotions, regulating them, and interpreting the emotions of others—is crucial for the development of healthy social interactions. Emotion understanding allows us to respond appropriately to others' needs, make predictions about social interactions, and even regulate our own emotions effectively. Research has shown that children who have better emotion understanding are rated as more socially skilled by their teachers, more likable by their peers, and are better able to navigate aggressive interactions. Likewise, children who have poor emotion understanding tend to be more aggressive, are more likely to develop behavioral problems and internalizing issues, and demonstrate lower academic achievement (Denham, 2019). But how do we develop emotion understanding? How do we learn to navigate our own emotions and the emotions of others?

There are various theories of what emotions are and how they come to be expressed and understood throughout human development. The dominant view in the social science literature has been that there are a set of basic discrete emotions—including anger, disgust, fear, happiness, sadness, and surprise—that are universally recognized and produced, regardless of individual experiences. According to this theory, basic emotions are innate, evolutionary adaptations to specific environmental challenges, and each has a distinct physiology and brain circuitry (Ekman & Cordaro, 2011; Izard, 2007; Panksepp, 2007). Evidence to support this view comes from classic research showing that individuals in an isolated, preliterate culture in New Guinea can perceive the emotional facial configurations of (White) American adults (Ekman et al., 1997). In this work, researchers showed participants photographs of White adults from the U.S. displaying various emotional facial configurations and asked them to match the configurations with emotional vignettes, such as “He is happy because his friends have arrived,” or “He is sad because his mother has passed away.” Despite having

no exposure to U.S. culture, participants from New Guinea successfully identified each emotion category.

These findings were robust, and subsequent studies using similar methods confirmed that there is significant cross-cultural agreement in recognizing emotion categories across different regions, including Asia, Europe, and the United States (Krause, 2004). This consistency has been found to extend to more subtle emotions, like pride (Tracy & Robins, 2008). Additionally, others have shown that people *express* emotions similarly across cultures as well, producing comparable emotional facial configurations in response to a range of emotional stimuli (Ghiselin et al., 1974; Scherer et al., 1988, 1986).

Because of the popularity of this view and its intuitive nature, many social scientists have ignored the role of development in emotion understanding. The discrete emotions perspective assumes that emotion perception and expression are universal, and thus, implicit in this perspective is the idea that emotion knowledge should develop rapidly early in life, with the ability to express the facial configurations of basic emotions like anger, sadness, and fear early in infancy, potentially at birth. This view also predicts that emotion knowledge develops on a similar trajectory across all individuals, relatively independent of context, culture, or experience.

However, the developmental literature is inconsistent with these assumptions. In contrast to emotions being early-emerging and unchanging over much of development, research has revealed a protracted trajectory for the development of emotional expression and perception, with emotion understanding accumulating incrementally over the course of infancy, early childhood, and even into adulthood and old age (Woodard et al., 2022). Broadly, this literature has suggested that early emotion understanding is very general, with both the expression and perception of emotions mapping roughly onto global positive and negative affect, and then gradually becoming more specific over time, with the ability to express and differentiate between more discrete emotions like anger, sadness, and fear (Widen, 2013; Widen & Russell, 2008, 2010).

Further, many researchers have reported significant variation in emotion perception and expression based on context and experience (Barrett et al., 2011; LoBue & Adolph, 2019). For instance, a particular facial configuration can be interpreted as disgusted in one context, such as when presented on a person who is eating, but can be interpreted as anger in another, such as when presented on a man making a fist (Aviezer et al., 2008). Additionally, not all emotional facial configurations are “universal” in how they are

interpreted. Research has shown that although emotions like happiness or sadness are cross-culturally recognized, other emotions—like achievement and relief—are only accurately recognized within specific cultural groups (Sauter et al., 2010). In fact, different cultures tend to focus on different cues for emotion recognition. Adults from Western countries, for example, rely heavily on facial configurations and verbal information, while adults from Japan prioritize contextual cues, and for example, are more likely to perceive facial configurations as happier when surrounded by other faces posing positive configurations (Ishii et al., 2003; Masuda et al., 2008; Tanaka et al., 2010).

Beyond variations in emotion perception, research has also highlighted cultural differences in emotional *expression*. For example, whereas adults from Western countries tend to express certain emotions through very distinct facial muscle configurations, individuals from East Asia show considerable overlap between categories of emotional facial configurations, especially for emotions like surprise, fear, disgust, and anger (Jack et al., 2014). Language also plays a crucial role in emotional expression, and the frequency with which people mention emotion-related words in response to different stimuli varies widely across cultures, even within Europe (Scherer et al., 1986). Furthermore, bilingual individuals express emotions differently depending on the language with which a stimulus is presented (Chen et al., 2012).

Differences in emotional expression between individuals have also been linked to societal norms. For instance, U.S. adults from low socio-economic backgrounds tend to express more anger than do individuals from higher socio-economic backgrounds, whereas the opposite pattern has been observed in Japan. This difference aligns with each culture's values: In Western cultures, independence and personal goal achievement are highly emphasized, and anger is often a response to blocked goals, particularly in lower socio-economic groups. In contrast, Japanese culture places a strong emphasis on social harmony and interdependence, discouraging anger in lower and middle socio-economic groups where it might disrupt group cohesion (Park et al., 2013).

As a result of these studies and many more showcasing the role of culture, context, language, and experience on emotion understanding, various alternative perspectives to the basic emotions approach have been proposed. We collectively call these views *process-based perspectives*. Although there are a variety of approaches that fit under this umbrella (including dimensional

models, constructivist approaches, and emergent theories), what they all have in common is the idea that emotions are learned through experience and the process of appraisal, highlighting the role of cognition as fundamental to both emotion perception and expression (Barrett, 2006; Clore & Ortony, 2008; Coan, 2010; Cunningham et al., 2013; Lazarus, 1984; Zajonc, 1980). In support of this view, one recent study demonstrated that when researchers gave over 300 people artificial intelligence (AI) tools to generate facial configurations that represent discrete emotions like happiness, fear, sadness, and anger, no two faces looked alike (Binetti et al., 2022).

Here, we will review the developmental literature on how infants and young children learn about emotions. We take a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. We will also emphasize the role of *input* into this development, a factor that has often been ignored. Indeed, little research has systematically sought to understand how children's emotional development is shaped by the emotional input they receive. This notion is central to a processed-based perspective.

We focus on the development of three abilities that comprise emotion understanding. First, we will discuss the developmental trajectory of *emotional expression*, and how infants and young children come to express emotions and behave emotionally. Second, we will discuss how infants and young children learn to *regulate* those emotional responses, and how different environmental inputs affect that ability. Third, we will discuss the development of *emotion perception*, or how infants and young children learn how to perceive, categorize, and identify the emotions of others over time. Finally, we will discuss mechanisms for developmental change in all of these domains, and how other concurrent domains of development—including motor skill, language, theory of mind, and attachment—might have cascading effects on changes in emotion understanding.

Altogether, taking a developmental perspective on how we learn to express, regulate, and perceive the emotions of others has implications for emotion theory, and the degree to which emotions are universally shared across individuals versus shaped by context and experience. Further, the work reviewed here has practical implications for how emotional input might shape emotion understanding in both adaptive and maladaptive ways. Thus, we will conclude by discussing how future research focusing

on how variability in the environment—or in the emotional input that infants and young children receive—shapes our emotional life for better or for worse.



1. Emotional expression

One way that children begin to develop emotion understanding is by experiencing emotions themselves and expressing them. There is some debate, consistent with the various theories of emotion discussed in the previous section, about whether newborn infants experience and express a vast array of emotions. According to the discrete emotions perspective, emotional expressions should be present at birth, and early in life emotional expressions should be identical to adult emotional expressions (Izard et al., 1987). Indeed, the very first studies in this area suggested that by 1 or 2 months of age U.S. infants can express facial configurations that represent fear, disgust, anger, sadness, happiness, and surprise (Izard et al., 1980). However, these findings were met with criticism, because, as often has been observed with newborns, spontaneous emotional facial configurations in very young infants were only produced very quickly and at random times. In other words, these early facial configurations were fleeting and not produced in response to an appropriate eliciting event, suggesting that they may not actually correspond to any underlying emotional state at all (Camras & Shutter, 2010). Further, in US samples, 4-month-old infants exhibit a combination of interest and surprise when confronted with tickling, tasting a sour substance, arm restraint, and the approach of a masked stranger. It is not until 12 months of age that US infants express clearly differentiated facial configurations of happiness, disgust, anger, and fear (respectively) in response to these events (Bennett et al., 2005).

In contrast, process-based views predict a more protracted developmental trajectory for the expression of specific emotions, with emotional expression co-developing with cognitive abilities. For example, happiness is a response to something rewarding and should only occur after infants have associated particular people or objects with a specific reward. Likewise, sadness is a response to the loss of a rewarding person or object and thus emerges with the ability to remember a rewarding person or object that is absent. Anger is a response to a blocked goal, requiring the ability to perform actions based on goals. Finally, fear is a response to a perceived threat, dependent on the ability to perceive a stimulus as a threat. Thus, according to a process-based perspective, infants would first have to experience or learn about rewarding

events (for happiness and sadness), engage in goal directed behavior (for anger), and recognize that a person or object is threatening (for fear) to experience and express each of these specific emotions.

In fact, infants do not express discrete happiness, or smiling, in response to a familiar person until 1 to 2 months of age, when they have developed the ability to recognize a familiar person. All prior smiling, or the neonatal smile, is thought to be a reflex, and is typically elicited during sleep, suggesting that even happiness is not reliably expressed without some cognitive prerequisite. Moreover, the emergence of the social smile, in contrast to being universal, appears to be experience dependent. Wörmann and colleagues, for example, demonstrated that the emergence of the social smile varies across cultures, with earlier emerging social smiles in cultures that practice more distal parenting characterized by face-to-face interactions (Kärtner et al., 2013; Wörmann et al., 2012, 2014).

Research has shown that the expression of other emotions emerges even later. For example, not until around 4 months of age do US infants exhibit discrete facial configurations of anger in response to an appropriate eliciting event—in most cases, in response to interrupting a learned contingency. In one classic study, researchers taught U.S. infants that whenever they pull on a string, an image of a smiling baby would appear on a screen. Infants were unable to learn this contingency until approximately 4 months. At that age, but not earlier, infants also showed evidence of a stereotypical angry facial configuration when the contingency was later interrupted (Lewis et al., 1990). This suggests that the ability to learn a contingency and then recognize when that contingency has been interrupted are cognitive prerequisites for the elicitation of discrete anger.

Similar findings have been reported on the development of the expression of fear. In the standard stranger approach paradigm, a stranger (typically a male adult) approaches an infant in one of many contexts, for example, while seated on the floor, in a highchair, or on their mothers' laps. Infants' behaviors—including facial configurations, crying, and escape behaviors—are measured. Importantly, although US 6-month-old infants can show that they differentiate between the faces of their mothers and strangers by looking longer at strangers, they do not respond differentially to the *approach* of a stranger until several months later (Bronson, 1972; Lewis et al., 1978; Lewis & Rosenblum, 1974; Sroufe, 1977). Thus, perceiving a face or individual as unfamiliar is not sufficient to elicit a fear response; infants do not show fear until they perceive the face of a stranger as a threat—an assessment that requires infants not only to recognize when someone

is novel versus familiar, but decide that the approach of the novel person is threatening or undesirable. Moreover, even when infants show fear in response to a stranger, typically between 8 and 12 months for US infants, classic studies have shown that the expression of fear in infants varies by context. US infants exhibit almost no evidence of fear when tested at home or when seated on a mother's lap, but they exhibit more negative emotional responses in the absence of their mothers, when tested in the lab, or when a stranger approaches rapidly or attempts to pick the infant up (Bronson, 1972; Ricciuti, 1974; Smith, 1974; Sroufe, 1977). This suggests that fearful responses towards strangers do not develop until about 8–12 months of age, and that context predicts these early responses.

The expression of more complex emotions also develops alongside cognitive abilities. The self-conscious emotions like embarrassment, envy, and empathy—which all require infants to evaluate their behaviors with reference to others' expectations or societal norms—do not appear until about 18 months of age, around the same time children begin to engage in self-referencing behavior; pride, shame, and guilt appear even later, around 36 months (Lewis, 2019). Further, once infants are old enough to reason about others' expectations, they might experience a more complex emotional response to an event that months earlier would have elicited a different response altogether. For example, researchers have reported that when 2-year-old US children experience failure, they are most likely to express sadness, whereas the very same failure in 4-year-old children is more likely to result in an expression of guilt (Lewis, 2019).

The trajectory for the development of disgust is even longer, despite assumptions that it is one of our most basic emotions and should be present at birth. Newborn infants are certainly capable of responding negatively to things that taste bad, but emotion researchers often call this *distaste*, which is a physiological response that requires gustatory contact (Rottman, 2014). Disgust may cause a physiological response, such as nausea, but requires no actual gustatory contact. In other words, in order to experience distaste, you have to taste something; disgust is more complex, and it does not require actually tasting anything. Indeed, disgust can be evoked not just by food, but also by blood, insects, or even by moral transgressions. Accordingly, while distaste is present at birth, disgust has a much more protracted period of development. For example, one classic study showed that U.S. children do not consistently reject a glass of juice that has been stirred with a used comb or that has a dead grasshopper visibly floating inside until about 6 to 8 years of age (Rozin et al., 1985). Although insects are not commonly eaten in the U.S. or Europe, grasshoppers, ants, and beetles are commonly consumed

in other parts of the world, such as in Mexico, Thailand, China, and parts of Africa and Australia. This suggests that disgust responses develop slowly, likely based on experience and cultural norms.

More recent work has highlighted that even within these emotion categories, there is also significant variation in emotional facial configurations across the lifespan. For example, in one study researchers coded 7- to 9-year-old children's facial configurations while they were having conversations with their mothers. Children consistently produced configurations of happiness during joyful conversations, but they produced anger, fear, and sadness more inconsistently; those configurations were not necessarily produced in conversations when they were discussing events that should have produced these emotions (Castro et al., 2018).

Furthermore, even when children do produce discrete emotional facial configurations in response to appropriate elicitors, these configurations do not necessarily look like the stereotypical configurations we often picture (and use in our research). According to the discrete emotions perspective, basic emotions should result in very specific facial configurations, such as smiling for happiness and scowling for anger. In fact, Ekman and colleagues have developed a system of coding emotional facial configurations called the Facial Affective Coding System (FACS) to reliably identify when a specific emotion is present via facial configuration. However, research with both adults (Durán & Fernández-Dols, 2021) and children (Camras et al., 2007a; Camras & Shutter, 2010; Castro et al., 2018) suggests that we express emotions with considerably more variability than the posed stereotyped configurations represent (Barrett et al., 2019; Durán & Fernández-Dols, 2021). For example, one study reported that US children between the ages of 3 and 6 rarely express the full stereotypical facial configurations for any emotion except happiness. When they express other emotions, some of the facial actions (e.g., shape of mouth, position of eyebrows) match the stereotypical configurations for those emotions, but rarely do all of the children's facial actions match the stereotypical configuration (Gaspar & Esteves, 2012).

The research discussed which has primarily been conducted with US children provides evidence that emotional expression develops slowly and reflects contextual factors. Comparison of developmental trajectories across cultures provides even stronger evidence of this conclusion. First, there is significant cultural variation in emotional expression across the lifespan. For instance, although Japanese and American adults tend to report feeling happiness, sadness, and anger, their experience of high arousal emotions (e.g., anger, fear, happiness) and low arousal emotions (e.g., sadness, calm)

differ. In Western or individualist cultures, high-arousal emotions are more valued, and adults thus report experiencing more high-arousal than low-arousal emotions. In contrast, in Eastern or collectivist cultures where low-arousal emotions are more valued, adults report experiencing more low-arousal emotions than high-arousal emotions (Lim, 2016).

Cross-cultural differences in emotional expression emerge even in infancy. As described earlier, the emergence of the social smile varies between cultures that differ in their use of distal parenting, which includes more face-to-face interactions (Kärtner et al., 2013; Wörmann et al., 2012, 2014). Infants across cultures also differ in the *intensity* of their emotional facial configurations. For example, one study showed that 12-month-old infants in the US were more intense in their expressions of pleasure and discomfort than were infants in Chile (Muzard et al., 2017). In another study of US, Japanese, and Chinese 11-month-old infants' responses to various emotion-eliciting activities in the lab, Chinese infants were less expressive, both smiling and crying less than US and Japanese infants (Camras et al., 1998). Likewise, Gartstein and colleagues reported that at 12 months, Spanish female infants showed more intense fear responses than did US infants (Gartstein et al., 2016).

Moreover, the development of these emotional expressions depends on experience, and there is evidence that there is more cross-cultural variability in emotions that requires some understanding of cultural expectations. In a large cross-cultural study of adults in China, India, Korea, Japan, and the United States, participants read brief stories that represented 22 emotions, and their emotional responses were coded. Interestingly, there was cross-cultural overlap (between 67–89 percent) for simpler emotions like surprise, fear, and anger, but very little correspondence (18–44 percent) for emotions like disgust and shame that have a longer developmental trajectory (Cordaro et al., 2018). Altogether, this work suggests that the developmental trajectory of emotional expression is closely tied to concurrent developments in cognition, with emotions that involve simpler cognitive processes, like happiness, developing sooner, and more complicated emotions like disgust and the self-conscious emotions developing later once children have some understanding of social norms.



2. Emotion regulation

Emotion regulation—or the ability to control one's emotions—similarly, and not surprisingly, has a protracted developmental trajectory. At

first, young infants are not able to regulate their emotional responses on their own and typically require caregivers to regulate their emotions for them. Techniques like rocking, swaddling, and most importantly touch, have been shown to be helpful in regulating infant negative affect, especially in newborns. For example, one group of researchers randomly assigned full term newborn infants in the Netherlands to receive either daily skin-to-skin contact or care-as-usual (which did not involve skin-to-skin contact) for 5 weeks. Infants who received the skin-to-skin contact cried less and slept more than infants in the control group. Further, there was a dose response effect, with more skin-to-skin contact leading to greater reductions in crying and longer total sleep duration (Cooijmans et al., 2022).

Further, skin-to-skin contact can have long-lasting effects on emotion regulation abilities, even into adulthood. For example, in another intervention study, a group of premature infants in Israel were assigned to either 2 weeks of skin-to-skin contact or standard incubator care. Infants who received the skin-to-skin contact demonstrated better emotion regulation 6 months later, and impressively, 10 years later. In fact, another study went on to report that US adults who were held and cuddled as infants were the most likely to be the most emotionally healthy and well-adjusted adults (Narvaez et al., 2016).

By 4 to 6 months of age, US infants begin to soothe themselves via touch, particularly at night, by stroking their own skin or petting a toy or stuffie. As they get older, these strategies become less physical and more cognitive, especially as children learn what emotions are acceptable and not acceptable to express. Importantly, there are different expectations about emotional expressivity within families, neighborhoods, and cultures, which emotion scientists have called *display rules*. As children develop, they learn the display rules specific to their environments and are tasked with regulating their emotions accordingly.

In a study of over 5000 people from 32 countries, researchers reported some consistency in display rules along with several cross-cultural differences. Across all the countries they tested, researchers reported that individuals thought it was more acceptable to be emotionally expressive with members of one's ingroup than one's outgroup. Further, certain negative emotions like contempt, disgust, and fear were the least acceptable emotions to express for all groups, and the expression of sadness was consistently associated with more acceptance in ingroups versus outgroups. However, there was cross-cultural variability as well. Western individualistic cultures accepted more expressiveness overall than Eastern collectivist cultures, especially for

positive emotions. Individualist cultures also had less variability in their regulation across emotions overall, likely because emotional expressiveness is generally more acceptable in these cultures, requiring less regulation.

Cross-cultural variation in the ability to regulate emotions is also evident in childhood. Conclusions about the development of emotion regulation are typically drawn from studies using Walter Michel's classic marshmallow task (or some variation) to test a related ability—delay of gratification. In the task, preschool-aged children are offered a marshmallow, or some equally appealing snack, but then are told that the experimenter needs to leave the room. The child is told that if they wait until the experimenter comes back, the child would be given two marshmallows to eat instead of just one. However, if the child is unable to wait, they can ring a bell to bid the experimenter's return and eat the single marshmallow instead of waiting for two. In the original study conducted in the US, children typically had difficulty waiting, particularly the youngest children (Mischel & Ebbesen, 1970).

Cross-cultural studies have suggested that there are cultural differences in children's ability to wait in this task, suggesting cultural differences in delay of gratification. In one study, researchers compared the behavior of 4-year-old children in Germany—a Western industrialized country—to Nso children, who are from an indigenous tribe in the Northwestern region of Cameroon. Like the US children tested in the original study, German children had difficulty waiting to eat the marshmallow (only 30 percent waited the entire time), and needed to use strategies to distract themselves, like sitting on their hands, talking to themselves, or looking away from the marshmallow. In contrast, the Nso children had little difficulty waiting for the second reward, with 70 percent waiting until the end. The researchers speculated that because children in the Nso culture are expected to control their negative emotions at a very early age, even young children were better able to regulate their emotional responses when faced with a challenge than German or US children who are more likely to be encouraged to express their negative emotions (Lamm et al., 2018). This suggests that children can regulate their emotions at a much earlier age in countries where the outward display of negative emotions is discouraged.

Other work suggests that delay of gratification may develop within cultural practices in which children are required to wait, even within cultures where outward displays of negative emotion are discouraged. In one study, preschool-aged children were tested in Japan and the US either on the standard marshmallow task or on a second similar task that involved waiting

to open a wrapped gift. In both cases, children could wait for two treats (two marshmallows or two gifts), or if they could ring the bell and have just one. Instead of global differences in the ability to wait, children from the two cultures could better regulate their emotions—and delay gratification—in different tasks. The US children had difficulty waiting in the marshmallow version of the task, but the Japanese children did not. However, the results were reversed for the wrapped gift—in this task, the Japanese children had difficulty waiting to open the gift, but the US children did not (Yanaoka et al., 2022). Importantly, these results reflect the norms of each country: In Japan, children are often expected to wait to eat lunch, for example, until everyone in their classroom is served—a norm that is not common in the US. In contrast, children in the US have experience waiting to open wrapped gifts, for example, by waiting until Christmas day to open presents that are visible under the Christmas tree or waiting until the end of a party to open a pile of wrapped presents on display. Children in Japan, in contrast, are often given gifts sporadically throughout the year instead of having specific events, and open those presents immediately. Altogether, this body of work suggests that like emotional expressiveness, emotion regulation is closely tied to concurrent developments in cognition, with significant cross-cultural variability especially with respect to culturally specific display rules.



3. Emotion perception and labeling

Finally, children's ability to perceive, understand, and talk about the emotions of others develops over time as well. Research examining US infants' perception of photographs of posed stereotypical emotional facial configurations suggest they have precocious abilities to discriminate such images, consistent with the discrete emotions perspective. Studies have provided evidence that even in the first few months of life infants discriminate between photographs of smiling (happy), frowning (sad), and wide-eyed (surprise) stereotypical facial configurations (Farroni et al., 2007; Field et al., 1983; Young-Browne et al., 1977). Further, around 5–6 months of age, they differentiate between wide-eyed gasping poses (fear), frowning poses (sad), and scowling poses (angry) (Schwartz et al., 1985; Serrano et al., 1992). By 6–7 months, infants group different models posing the same emotion category (C. A. Nelson et al., 1979) and detect category boundaries when images were morphed from one stereotypical configuration to another (Kotsoni et al., 2001). These findings suggest that infants' ability to

differentiate between a wide variety of emotion categories develops rapidly in the first year of life.

However, other studies have suggested a more protracted developmental trajectory for emotion perception. In one study, US 3- to 6-year-old children and adults were asked to sort images of emotional facial configurations according to whether they were related. The faces differed in valence, with some positive and some negative, and they represented several distinct emotions (e.g., happiness, fear). In general, across age, children did not become more competent with this task, but rather younger and older children used different information to sort the faces. The younger children relied on valence, whereas it was not until 5 years of age that children consistently sorted the images based on discrete emotion categories (Woodard et al., 2022). Results like these suggest that although infants are sensitive to perceptual differences between facial configurations, children's understanding of the emotional expressions of others continues to develop into childhood.

Furthermore, although infants can discriminate between faces with different emotional facial configurations, it is impossible to know from most studies whether infants are simply discriminating between the perceptual features of various emotional facial configurations or whether they can discriminate between the emotional meaning that these configurations represent. It is not until infants can use emotional facial configurations to guide their action that we can know for sure that they are interpreting meaning from these configurations. The first evidence of such a process is the emergence of *social referencing* late in the first post-natal year in US infants (Sorce et al., 1985). When infants experience an ambiguous situation, they look to a caregiver or trusted adults' facial configuration and modify their behavior based on the configuration they see.

For example, at 12 months of age, infants look to their mothers' facial configurations to guide behavior when presented with an ambiguous visual drop-off (Sorce et al., 1985) or slope (Tamis-LeMonda et al., 2008). Infants are much more likely to traverse across the drop-off or down the slope if their mothers have a positive emotional facial configuration than if they have a negative configuration, for example conveying fear. Infants behave similarly when presented with novel objects: US 12-month-old infants avoid touching novel objects that are paired with a fearful facial configuration compared to novel objects paired with a happy or neutral configuration (Mumme & Fernald, 2003). Similarly, after mothers pose either a positive or negative facial configuration towards one of two novel toys, toddlers

are more likely to avoid touching the toys that are paired with negative configurations than toys that are paired with positive configurations (Gerull & Rapee, 2002). Such results suggest that late in infancy, US children have begun to associate facial configurations with different meaning, and use those differences to determine how to behave, particularly when faced with novel or ambiguous situations.

As further evidence of developing emotion perception, between the ages of 2 and 3 children begin to produce the expected labels for various stereotyped facial configurations, first by correctly distinguishing smiling (positive) versus scowling or frowning (negative) poses, then later, by correctly producing the labels for various emotion categories (Widen, 2013; Widen & Russell, 2008). Importantly, in most of these studies the stimulus sets of emotional facial configurations are high in emotional intensity and represent only the most stereotypical of emotional facial configurations. Thus, the results may not generalize to the more varied and subtle facial configurations children encounter in their everyday lives. A handful of studies using facial configurations of more varied intensities suggest a very protracted development in children's ability to accurately identify and label low intensity configurations. Indeed, although children between the ages of 7 and 10 are highly accurate at identifying high intensity emotional facial configurations, it is not until much later that children accurately identify lower intensity configurations. Moreover, this developmental trajectory differs for different categories of emotion (e.g., happy versus disgusted). In general, this research suggests that the perception and labeling of emotional facial configurations does not reach maturity until adulthood (Gao & Maurer, 2010; Thomas et al., 2007).



4. Mechanisms and cascades

Thus far, we have discussed the trajectory for the development of emotion understanding, which includes the development of emotional behavior, emotion regulation, and emotion perception, over the first several years of life. But what are the mechanisms by which infants acquire emotion knowledge in these domains? Because much of the literature is based on the discrete emotions perspective, there is remarkably little research on identifying the mechanisms by which emotional understanding develops. That is, there is a large literature documenting when children of different ages perceive, identify, and understand something about different facial configurations, as well as how children express and regulate their own

emotions, but there is less work attempting to understand how these abilities develop.

We can think about the development of emotion understanding by considering emotions as part of a system, with development reflecting cascading interactions across multiple domains. As part of this approach, it is important to recognize that emotion understanding is not just one thing. As described above, emotion understanding involves *emotional expression* and the *regulation* of that expression, often requiring the internalization of cultural norms. Emotion understanding also involves *perceiving* and *recognizing* the emotions of others, typically relying on emotional facial configurations, but there are many cues to a person's emotional state. Finally, emotion understanding involves verbalizing one's own and other's emotional state. Thus, it is too simplistic to think about the development of emotion understanding as one thing; instead, as described here, there are various developmental trajectories for different aspects of emotion understanding, and those trajectories differ for different emotions.

Furthermore, the development of emotion understanding does not happen in isolation but rather changes with development in other systems. As described here, emotion understanding is closely tied to developmental changes in cognitive processes, social interactions, and learning of cultural norms and expectations. It is impossible to understand the development of emotion understanding without recognizing how changes in the aspects of emotion reflect changes in other domains.

4.1 The role of input

There are several mechanisms by which infants and young children acquire emotion knowledge, from general purpose mechanisms like statistical learning, to the transmission of facial and verbal emotional information by observation or interacting with others. All these mechanisms involve the child processing and evaluating their experiences and input, however little is known about the kind of emotion input children actually receive. Focusing on the role of input has long been important in some areas of development, for example in how children acquire language (Hoff, 2003; Huttenlocher et al., 1991; Rowe & Snow, 2020) or how they form attachments to their caregivers (Bowlby, 1979). Increasingly, researchers have begun to consider how the nature of the input shapes other aspects of development, such as infants' emerging understanding of objects and object categories (Smith et al., 2018).

Input comes from interactions with others and what others model. Infants presumably learn about emotional expressions by observing the emotional expressions of others. This is evidenced by the fact that cultural differences in emotional expression are observed even in infancy (Camras et al., 2007b, 1998). Similarly, children receive input through *social referencing* or looking to others for emotion information about something or someone that is otherwise ambiguous. As mentioned above, by 12–18 months of age, US infants use the emotional facial configurations of others to guide their behavior, suggesting not only that infants recognize those facial configurations as emotion related, but also that they use them to learn about the relevant emotions in ambiguous situations.

Experimental studies have shown that exposure to specific emotional facial configurations leads to specific kinds of emotion learning. For example, children between the ages of 7 and 10 years are slower to approach novel animals that were previously paired with a fearful versus happy facial configuration. Most importantly, children also report a higher rate of fear about the animals that were paired with a fearful facial configuration when compared to those paired with a happy configuration (Askew & Field, 2007; Broeren et al., 2011). Further work has shown that these effects—namely increased fear—can last up to 6 months after initial exposure (Field et al., 2008; Muris et al., 2003), suggesting that information gleaned by social referencing can not only influence infants' behavior in the moment, but can also shape children's longer-term emotional responses (Dubi et al., 2008; Gerull & Rapee, 2002).

Children also learn from the emotional language they hear. For example, 6- to 9-year-old children show increased heart rate and slower approach responses to novel animals after being presented with negative versus positive or neutral verbal information (Field & Lawson, 2003; Field & Schorah, 2007). Likewise, US 5-year-old children report more fear of a stranger after hearing parents provide negative verbal information about the stranger (Aktar et al., 2022). At younger ages, exposure to labels can aid infants in forming categories of emotions (Price et al., 2022; Ruba et al., 2020), just as labels aid infants in forming categories of spatial relations (Casasola, 2005).

Children can also build their emotion knowledge through parent-child conversations. For example, mothers' use of language about desire predicts their 15-month-old toddlers emotion understanding when the children were 24 months (Taumoepeau & Ruffman, 2006, 2008). Similarly, in a study where US children and parents discussed a wordless storybook with images of actors experiencing various emotions, parents' questions and emotion talk

were related to children's emotion talk over time (Reschke et al., 2024). This suggests that emotion language can serve as input to help children understand their own emotional states and how to express their emotions in culturally appropriate ways.

Children also develop emotion understanding from detecting statistical regularities in the facial configurations they observe. For example, children may detect statistical regularities in the input and use statistical learning to identify emotion boundaries. A large number of studies have shown that infants have sophisticated statistical learning abilities to identify and categorize sounds in speech (Saffran et al., 1996) and objects in visual stimuli (Needham et al., 2005). It is therefore plausible that infants also detect statistical regularities in facial configurations and use those regularities to identify emotion boundaries.

In fact, one study has shown that US 6- to 8-year-old children and adults categorize emotional facial configurations using statistical regularities. In this study, after learning how to categorize morphed faces as angry or neutral (in reality each face was created by morphing an image of an angry face and an image of a neutral face), participants shifted their category boundaries when they experienced a change in the distribution of angry and neutral faces (Plate et al., 2019). This finding demonstrates that at least by 6 years of age, children are sensitive to the distribution of facial configuration features in their experience, and their emotion boundaries reflect that distribution.

Much less is known about whether *infants* use statistical learning to recognize emotion boundaries. One problem is that despite a few studies examining infants' visual input for the presence of faces (Jayaraman et al., 2015, 2017; Sugden et al., 2014; Sugden & Moulson, 2017), we know very little about what facial configurations infants actually see. However, research with clinical populations suggests that infants do learn the statistical regularities in their input. For example, infants of depressed or anxious mothers show different perception of and reactions to emotional facial configurations in the lab. Depressed mothers express more negative affect than do non-depressed mothers (Murray et al., 1993), even in face-to-face interactions with their infants (Aktar et al., 2017; Feldman et al., 2010). Accordingly, infants of depressed mothers—who presumably had more negative emotion input than infants of non-depressed mothers—show less interest in negative facial configurations than infants of nondepressed mothers (Cohn et al., 1986; Field, 1992), and they take longer to disengage or look away from stereotypically happy configurations (Diego et al., 2004; Field et al., 1998). This result may reflect the same processes shown by Plate

et al. for older children; these infants may have a shifted emotion boundary reflecting the distribution of negative and positive emotion input in their experience. As a result, positive facial configurations may be more novel to infants of depressed mothers, and these differences may reflect infants of depressed mothers showing a novelty response to positive configurations and a habituation response to negative configurations.

Similar patterns have been observed with infants of anxious mothers. Research suggests that anxious mothers are less sensitive to their infants' needs, less emotionally responsive, and more interfering when compared to non-anxious mothers (Kaitz & Maytal, 2005). Infants of anxious mothers have difficulty disengaging or looking away from negative facial configurations, especially angry or threatening stereotypes, when compared to infants of non-anxious mothers (Morales et al., 2017). Further, longitudinal work has shown that infants of anxious mothers engage less and become slower to detect emotional facial configurations overall over the course of the first two years of life (Vallorani et al., 2024). Similar to infants of depressed mothers, infants of anxious mothers may have formed different emotion boundaries reflecting the distribution of the positive and negative facial configurations in their experience.

4.2 Cascades across domains

As described earlier, emotion understanding does not develop in isolation. Instead, developments in other domains have cascading effects on emotion understanding over time, exposing children to new kinds of input and experiences. Consider, for example, developmental changes in infancy in basic cognitive abilities such as perception, attention, and memory. Changes in each of these abilities affect the developing ability to reason about emotions (Ruba & Pollak, 2020). For example, at birth, infants have extremely poor visual acuity and immature visual attention and perception. Nonetheless, some studies suggest that newborn infants distinguish between faces with different emotional facial configurations (Farroni et al., 2007; Field et al., 1983; Young-Browne et al., 1977). Given their limited visual abilities, any such discrimination between different emotional facial configurations cannot be due to attention to, perception of, and recognition of the details of the facial configurations associated with those emotional expressions. However, the discriminations young infants make based on their limited visual experience may be a critically important step in the development of emotion understanding. Indeed, research with individuals born with dense

central cataracts, and therefore are essentially blind from birth, suggests that early visual experience is critically important for the development of typical face perception (Le Grand et al., 2001). This early visual experience may also be important in the development of visual perception of emotions.

This is just one example of how developmental changes in basic cognitive abilities might have a cascading effect on the development of emotional understanding. As infants' visual perception abilities develop, they can actually detect the features of facial configurations and differentiate them from one another, allowing categorization based on more sophisticated and subtle features and configurations. Infants must learn to attend to the most relevant features and combine information stored in memory from different encounters. Thus, the development of emotion understanding is closely linked to the development of core cognitive abilities.

Second, language development contributes to emotional development. For example, the ability to perceive and learn emotion labels may help young children recognize emotion boundaries. Research with US infants suggests that at the end of the first year and early in the second year of life, the presence of labels facilitates their attention to other kinds of categories (e.g., horses or purple things) (Forbes & Plunkett, 2019; Fulkerson & Waxman, 2007; Waxman & Booth, 2003). Thus, it is possible that young children's learning of emotion categories is influenced by their developing ability to recognize labels as markers of abstract categories (Hoemann et al., 2020). By the age of 2 or 3, US children do use emotion labels in this way. Studies have demonstrated that toddlers in the second year can associate some emotion labels with visual stimuli (Ruba et al., 2019) and at least by age 2 or 3, being given labels seems to help children categorize some emotional expressions (Price et al., 2022). Thus, young children's developing ability to use labels to organize visual stimuli into categories likely influences how they categorize facial configurations into emotion categories.

Further, as described earlier, language is one way parents might transmit emotion knowledge to their children. An analysis of parent-child interactions in the CHILDES database (for dyads speaking North American English or British English) showed that parents used emotionally valenced words in the utterances surrounding emotion words (Nencheva et al., 2023). For example, when using words with their child such as "happy" or "sad," parents' other utterances near those words were positive (for happy) or negative (for sad).

Motor development may also contribute to changes in emotion understanding. For example, before infants can sit independently, they likely

spend a great deal of time laying on their backs, mostly looking at other people's faces. Indeed, there is significant exposure to facial information early in development, but this exposure gradually declines as infants begin to sit on their own and shift their focus to handling objects (Fausey et al., 2016; Sugden et al., 2014; Sugden & Moulson, 2019). The relatively high exposure to face perception early in infancy may similarly be important for infants' developing perception, recognition, and understanding of facial configurations. That is, early motor abilities (or general lack thereof) may result in more face exposure, providing infants with critical visual experience they need for not only developing face perception but also developing emotion understanding.

As infants get older and develop the abilities to sit, crawl, and walk, there are fewer faces in their field of view (Franchak et al., 2018) and they must shift to other sources of input (e.g., language, prosody, context) to learn about emotions. Furthermore, it has been suggested that once infants are able to locomote, either by crawling or walking independently, this independence might cause increased bouts of negative affect from caregivers, as infants can, for the first time, defy the rules of the house (Campos et al., 2000). Altogether, this suggests that changes in motor ability could readily cascade into important changes in the content and type of emotional input that infants receive.

Development of children's theory of mind, or the ability to reason about the mental states of others, between the ages of 2.5 and 5 years of age, is another system that contributes to emotional development. As mentioned above, the self-conscious emotions like embarrassment, envy, and empathy all require children to evaluate their behaviors with respect to others' expectations. Not surprisingly, there is research showing a causal link between experiencing the self-conscious emotions and training in theory of mind. In one study, US elementary school children were given training in acting, which requires children to put themselves in the mindset of another person. Children who were given the acting training showed enhanced theory of mind and empathy one year later when compared to a control group that received visual arts or music training (Goldstein & Winner, 2012). Thus, advances in children's ability to reason about the minds of others can cascade into changes in children's ability to experience self-conscious emotions, and to more accurately reason about the emotions of others.

Finally, attachment quality, or how secure an infant feels in the presence of their primary caregivers, can have cascading effects on emotion perception and regulation. Using event related potentials (ERP), researchers have

shown that although children in institutionalized care have greater neural responses to fearful emotional faces than children in a non-institutionalized comparison group, this difference disappears when children establish a secure attachment relationship with a foster parent (Nelson et al., 2013; Young et al., 2017). This suggests that feeling securely attached to a caregiver may buffer infants' responses to certain negative or threatening facial configurations so that these configurations are perceived as less threatening.



5. Conclusions and future directions

Here, we reviewed the developmental literature on how infants and young children learn about emotions based on a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. As such, the work reviewed here moves us beyond traditional emotion perspectives by highlighting how variation in input shapes the developmental trajectory of emotion knowledge over time. This trajectory may appear abrupt at first blush, suggesting that infants learn basic emotion categories in the first year of life, but the evidence we discussed points to a much more protracted developmental trajectory, with significant change across childhood.

However, additional future research that focuses on how specific input drives emotion understanding is still needed. Recent work suggests that even brief changes in social interactions can have a lasting effect on emotion and mental state knowledge. For example, one very recent study examined false belief understanding in two cohorts of US 3- to 5-year-old children who were either tested before or after the COVID-19 lockdown in 2020. They reported that children tested after lockdown had significantly worse false-belief knowledge than children tested before (Scott et al., 2024). This suggests that there could be significant cohort effects based on global events that encourage or suppress social interaction, which can potentially in turn, have an effect on emotion understanding.

This is important, as changes in emotion understanding can shape children's long-term mental health. For example, we discussed here how infants of anxious mothers have difficulty disengaging or looking away from negative facial configurations, especially angry or threatening stereotypes, when compared to infants of non-anxious mothers (Morales et al., 2017). Difficulty disengaging from emotional stimuli has been associated with the

development of anxiety in infants and children. Thus, emotional input from depressed or anxious caregivers might set the stage for the development of these same emotional problems in the infants themselves (Burris et al., 2019). This suggests that the emotional input children receive can put them at risk for various emotional problems later in life, making further research on this topic important for potentially mitigating the development of maladaptive patterns of emotion reasoning.

Finally, we made the case in this chapter for the importance of input in building emotion knowledge. We provided examples from studies of cross-cultural and research from clinical populations. However, to date, we still know very little about the *natural* emotional input that infants typically receive in the first few years of life, at a crucial time when infants are first learning to recognize and differentiate between emotion categories. As discussed above, most of the research in this domain focuses on how infants and young children perceive and label photographs of posed, stereotypical emotional facial configurations. This is problematic for two reasons. First, as mentioned above, by the time infants reach 6 months of age, they experience a sharp decline in their attention to faces in general, likely because of the shift from predominantly laying on their backs to sitting independently. Thus, the reliance only on emotional information from static posed facial configurations is not necessarily representative of infants' daily experience.

Second, in the real world, adults rarely express instances of emotion using posed stereotypical configurations (Barrett et al., 2019), so it is likely that the input infants receive is also highly variable and context-dependent (Barrett et al., 2019; Hoemann et al., 2020; LoBue & Adolph, 2019). However, scientists have very little information about what emotional expressions infants actually see, and therefore what real-world input they rely on to develop emotion understanding. So while the existing developmental literature can tell us about infants' increasing ability to detect and classify stereotypical, static facial configurations, it might mischaracterize the real-world learning problem infants face altogether. In other words, rather than learning relatively consistent facial configurations that are centered around a stereotype, infants must learn to make inferences about dynamically changing facial and vocal expressions that are fully contextually situated. This is likely a good thing, as researchers have argued that the best way to improve learning and generalization is to expose the learners to more variable input (Raviv et al., 2022). But if this is indeed the case, we still have a great deal to learn about how infants build emotion knowledge in real world contexts. These data are vital for discovering how infants infer the

emotions of others, build expectations about social interactions, plan their behavior adaptively, and how specific experiences and individual differences shape emotional development (Hoemann et al., 2020). Further, they are *absolutely critical* for discovering how atypical emotional environments, such as those associated with maltreatment, anxiety, or depression, might shape maladaptive trajectories of emotional behavior. We hope this review helps draw attention to the importance of input in the development of emotion knowledge, and in turn drives future research in the field with this important focus.

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